

9(a)	piezo-electric crystal	B1
	(ultrasound) wave causes shape change / vibrations (of crystal)	B1
	shape change / vibrations causes e.m.f. (which is detected)	B1
9(b)(i)	93V	A1
9(b)(ii)	$2.7 \times 10^7 \text{ rad s}^{-1}$	A1
9(c)(i)	$\text{kg m}^{-2} \text{ s}^{-1}$	B1
9(c)(ii)	$\rho = Z / c = 1.7 \times 10^6 / 1600$ $= 1100 \text{ kg m}^{-3}$	A1
9(c)(iii)	intensity reflection coefficient ≈ 1 or Z_1 and Z_2 are very different	B1
	almost no / no ultrasound transmitted (into air filled cavity)	B1